

Rice Leaf Disease Classification

DeiT-Small | Canonical Dataset V1

Final validation evaluation after training on the canonical stratified 80/10/10 dataset.

1. Experiment Configuration

Parameter	Value
Model	deit_small_patch16_224
Image Size	224 × 224
Batch Size	32
Epochs	25
Learning Rate	0.0001
Weight Decay	0.01
Optimizer	AdamW
Scheduler	CosineAnnealingLR
Random Seed	42
Device	cuda
GPU	Tesla T4
Mixed Precision	True

2. Canonical Dataset

Split	Images	Usage
Train	3063	Model training
Validation	383	Checkpoint selection / evaluation
Test	383	LOCKED. Not used in this experiment.

Canonical dataset path: /kaggle/input/datasets/varun2ks05/newdata

3. Final Validation Metrics

Metric	Result
Best Validation Accuracy	99.22%
Best Epoch	17
Final Validation Accuracy	99.22%
Macro Precision	99.22%
Macro Recall	99.21%
Macro F1-Score	99.21%

4. Classification Report

precision recall f1-score support

Bacterial Leaf Blight 1.0000 0.9683 0.9839 63

Brown Spot 1.0000 1.0000 1.0000 65

Healthy Rice Leaf 1.0000 1.0000 1.0000 66

Leaf Blast 0.9844 1.0000 0.9921 63

Leaf scald 0.9688 0.9841 0.9764 63

Sheath Blight 1.0000 1.0000 1.0000 63

accuracy 0.9922 383

macro avg 0.9922 0.9921 0.9921 383

weighted avg 0.9923 0.9922 0.9922 383

5. Confusion Matrix

True \ Pred	0	1	2	3	4	5
0	61	0	0	0	2	0
1	0	65	0	0	0	0
2	0	0	66	0	0	0
3	0	0	0	63	0	0
4	0	0	0	1	62	0
5	0	0	0	0	0	63

Class index mapping:

0 = Bacterial Leaf Blight

1 = Brown Spot

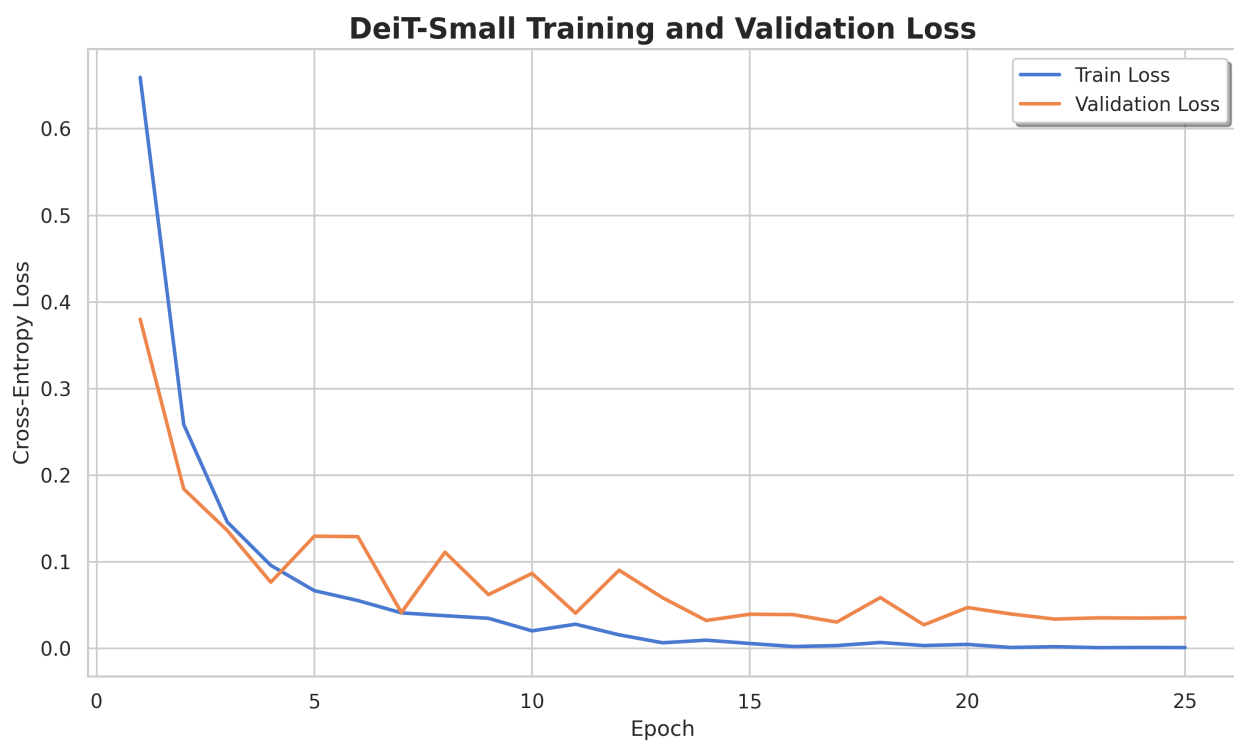
2 = Healthy Rice Leaf

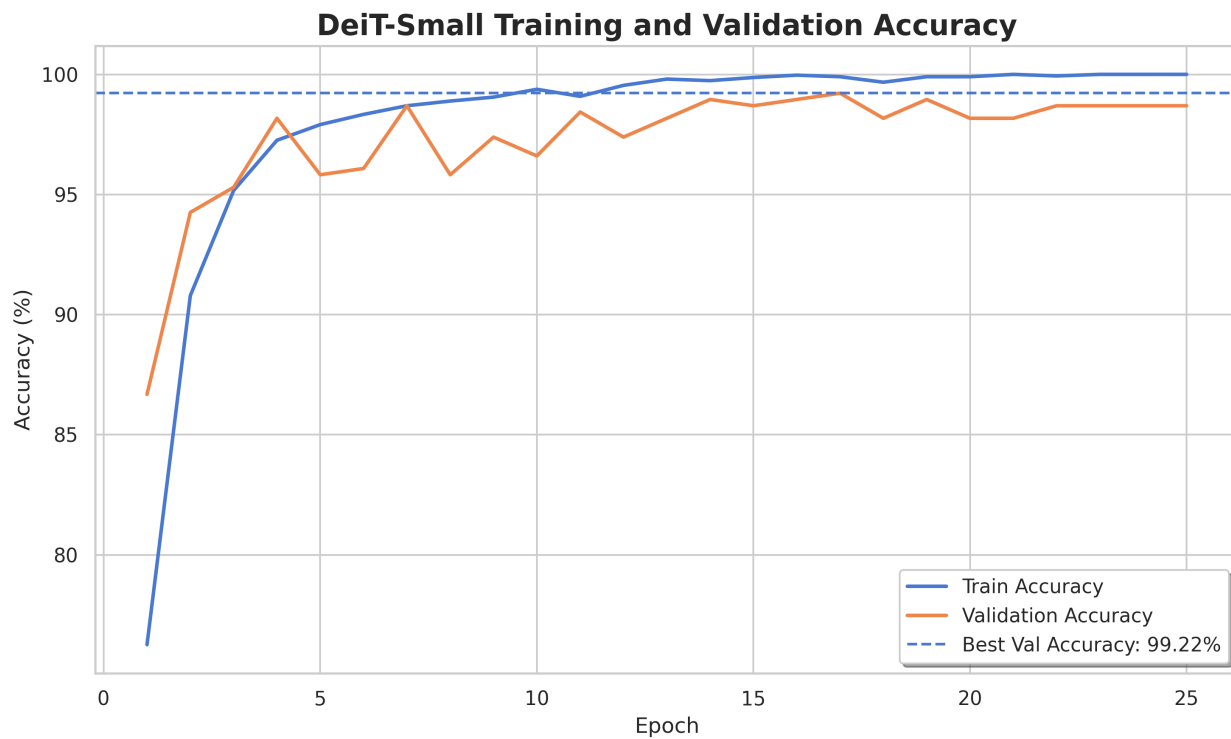
3 = Leaf Blast

4 = Leaf scald

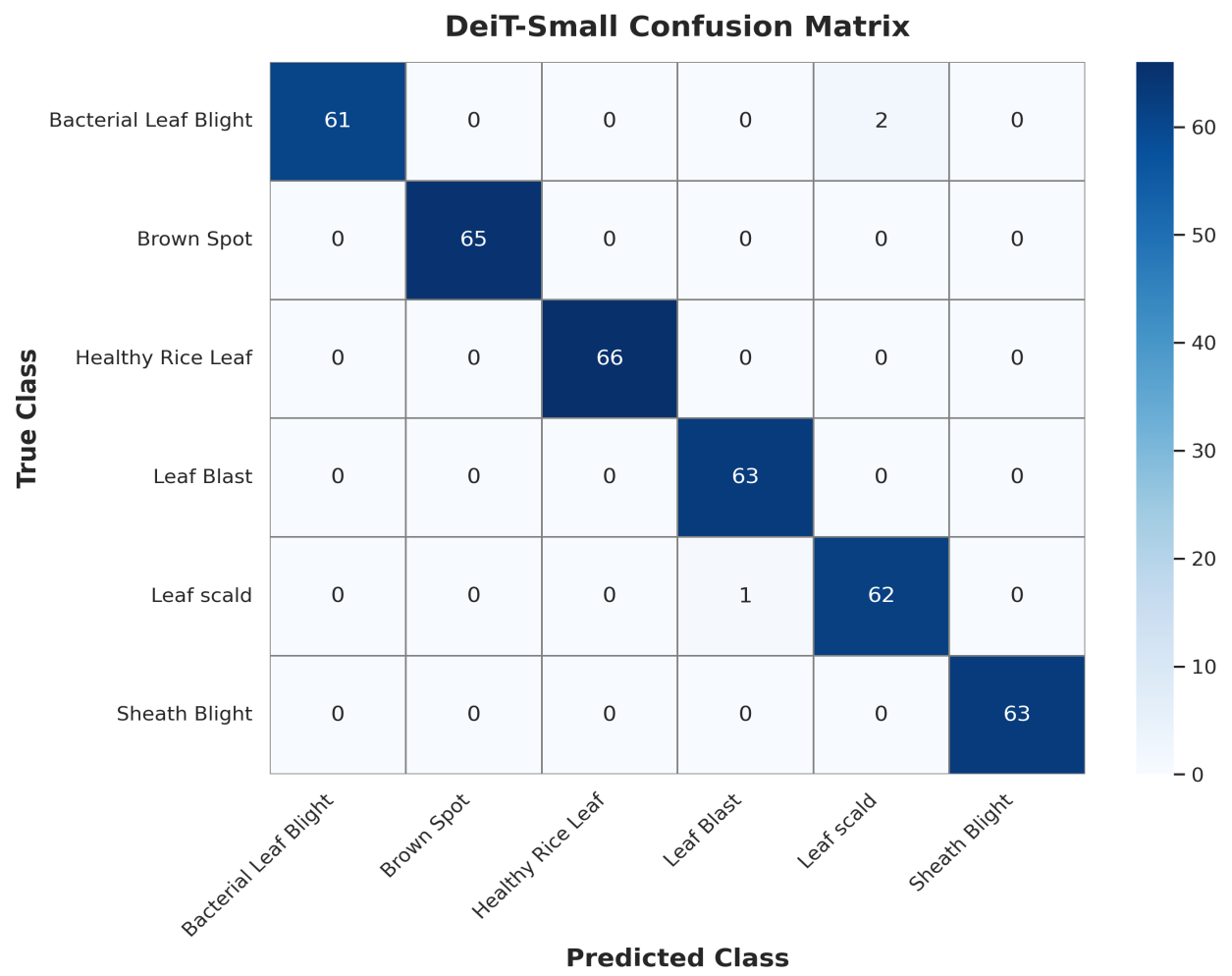
5 = Sheath Blight

6. Training Curves

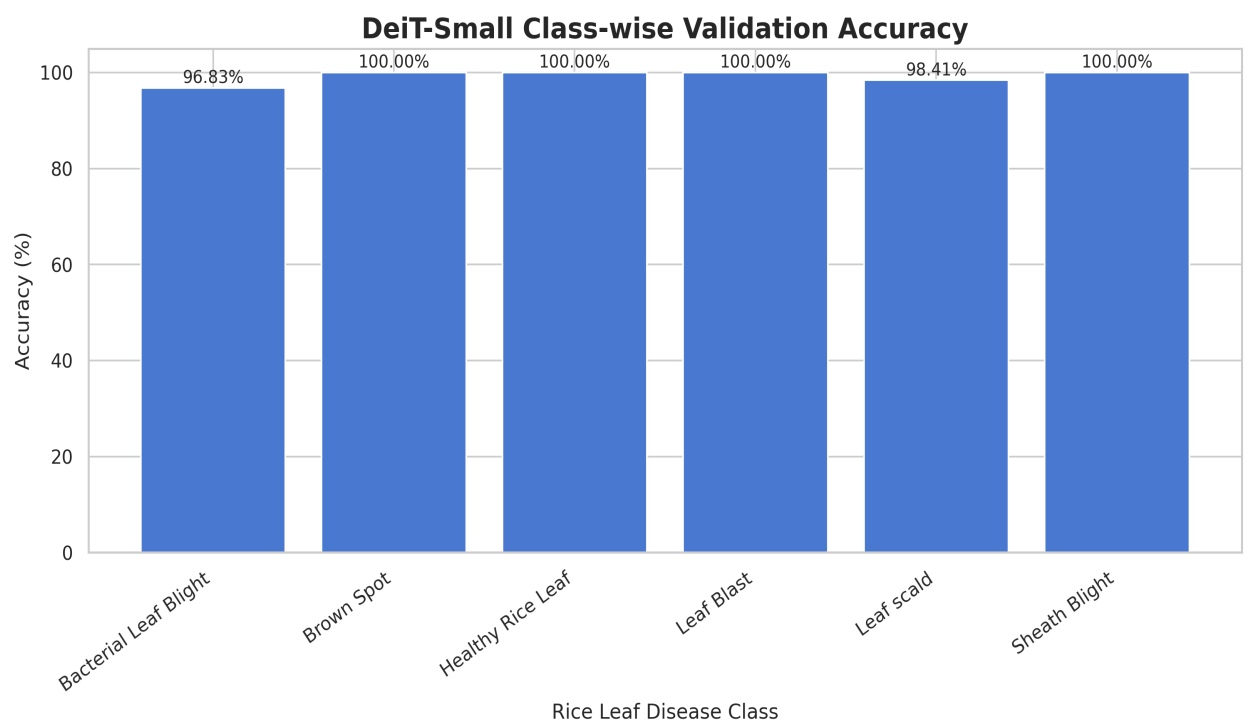




7. Confusion Matrix Visualization



8. Class-wise Accuracy



9. Complete Epoch History

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	LR
1	0.6593	76.27%	0.3802	86.68%	9.96e-05
2	0.2587	90.79%	0.1842	94.26%	9.84e-05
3	0.1461	95.17%	0.1364	95.30%	9.65e-05
4	0.0961	97.26%	0.0766	98.17%	9.39e-05
5	0.0669	97.91%	0.1297	95.82%	9.05e-05
6	0.0555	98.33%	0.1292	96.08%	8.66e-05
7	0.0412	98.69%	0.0416	98.69%	8.21e-05
8	0.0380	98.89%	0.1113	95.82%	7.70e-05
9	0.0350	99.05%	0.0623	97.39%	7.16e-05
10	0.0205	99.38%	0.0868	96.61%	6.58e-05
11	0.0282	99.09%	0.0409	98.43%	5.98e-05
12	0.0159	99.54%	0.0905	97.39%	5.36e-05
13	0.0067	99.80%	0.0587	98.17%	4.74e-05
14	0.0097	99.74%	0.0324	98.96%	4.12e-05
15	0.0059	99.87%	0.0397	98.69%	3.52e-05
16	0.0024	99.97%	0.0392	98.96%	2.94e-05
17	0.0035	99.90%	0.0306	99.22%	2.40e-05
18	0.0071	99.67%	0.0589	98.17%	1.89e-05
19	0.0035	99.90%	0.0274	98.96%	1.44e-05
20	0.0048	99.90%	0.0474	98.17%	1.05e-05
21	0.0013	100.00%	0.0400	98.17%	7.12e-06
22	0.0022	99.93%	0.0341	98.69%	4.48e-06
23	0.0011	100.00%	0.0355	98.69%	2.56e-06
24	0.0013	100.00%	0.0352	98.69%	1.39e-06
25	0.0012	100.00%	0.0356	98.69%	1.00e-06

10. Experiment Notes

- The dataset uses the canonical stratified 80/10/10 split.
- Random seed: 42.
- The training and validation sets were used for model development.
- The test set was intentionally kept completely untouched.
- The best model was selected using validation accuracy.
- The saved checkpoint corresponds to the epoch with the highest validation accuracy.
- The test set should be evaluated only during the final locked evaluation stage, after all individual models and ensemble decisions have been finalized.